Let one side of a rectangle be $(x+a)$ and the other side be $(x+b)$ units.

Then, the total area of the rectangle ABCD=length ×breadth=$(x+a)(x+b)$ ...(3)

From the Fig. 3.10, we can see that the

Area of the rectangle ABCD = area of the square AFIE + area of the rectangle FBGI + area of the rectangle EIHD + area of the rectangle IGCH

$$= x^2 + ax + bx + ab$$
$$= x^2 + x(a+b) + ab \quad ...(4)$$

From (3) and (4) we get, $(x+a)(x+b) = x^2 + x(a+b) + ab$

Hence, $(x+a)(x+b) = x^2 + x(a+b) + ab$ is an identity.

Note

(i) In case if $b=-b$ then the identity is $(x+a)(x+(-b)) = x^2 + x(a+(-b)) + a(-b)$

$$(x+a)(x-b) = x^2 + x(a-b) - ab$$

(ii) If $a=-a$ then the identity is $(x+(-a))(x+b) = x^2 + x((-a)+b) + (-a)b$

$$(x-a)(x+b) = x^2 + x(b-a) - ab$$

(iii) If $a=-a$ *and* $b=-b$ then the identity is

$$(x+(-a))(x+(-b)) = x^2 + x((-a)+(-b)) + (-a)(-b)$$
$$(x-a)(x-b) = x^2 - x(a+b) + ab$$

Example 3.1 Simplify the following using the identity $(x+a)(x+b) = x^2 + x(a+b) + ab$:

(i) $(x+3)(x+5)$ (ii) $(y+8)(y+6)$ (iii) 43 x 36

Solution

(i) $(x+3)(x+5)$

Let us represent the expression geometrically, as shown in the *Fig. 3.11*.

In the rectangle with length $(x+5)$ and breadth $(x+3)$, we get,

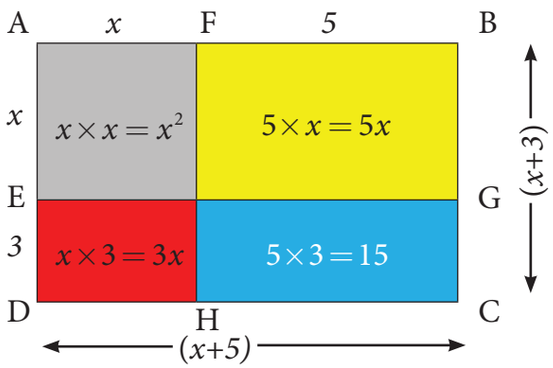


Fig. 3.11

Area of bigger rectangle = Area of a square + Area of three rectangles

Therefore,

$$(x+3)(x+5) = x^2 + 5x + 3x + 15$$
$$= x^2 + (5+3)x + 15$$
$$= x^2 + 8x + 15.$$

(ii) $(y+6)(y+8)$

Let us represent the expression geometrically, as shown in *Fig. 3.12*.

In the rectangle with length $(y+6)$ and breadth $(y+8)$ units, we get,

Area of bigger rectangle = Area of square + Area of three rectangles

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